

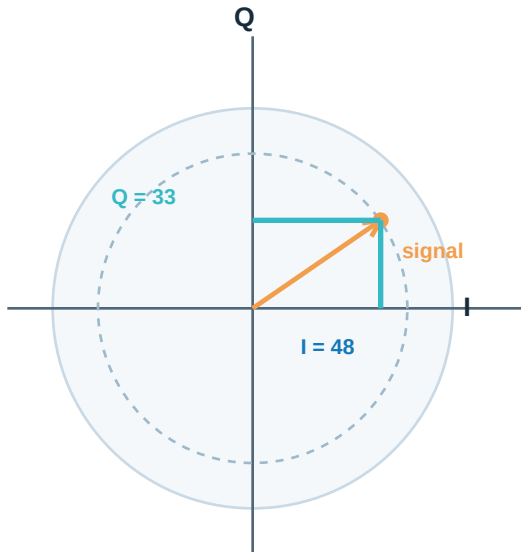
How SDR hears radio with I/Q

Two number streams preserve amplitude and phase - enough to reconstruct the signal in software.

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1. A radio wave is a rapidly spinning vector

At 100 MHz, the voltage from an antenna changes 100 million times each second. An SDR shifts the station near zero frequency, then measures its two perpendicular components.



Think of I and Q as coordinates

I (in-phase) is the projection onto a cosine reference. Q (quadrature) is the projection onto a sine reference, 90 degrees away. Together they describe one instantaneous point in the complex plane:

$$z[n] = I[n] + j Q[n]$$

magnitude = strength | angle = phase

2. What the SDR hardware actually does



A local oscillator is mixed with the chosen RF signal. One copy uses cosine, the other sine. Low-pass filters remove the high-frequency sum, leaving the slow I and Q differences.

WHY TWO STREAMS?

A single real stream cannot distinguish a positive frequency from its mirror image. I/Q retains rotation direction, so filters and demodulators can separate nearby signals cleanly.

WHAT A SAMPLE MEANS

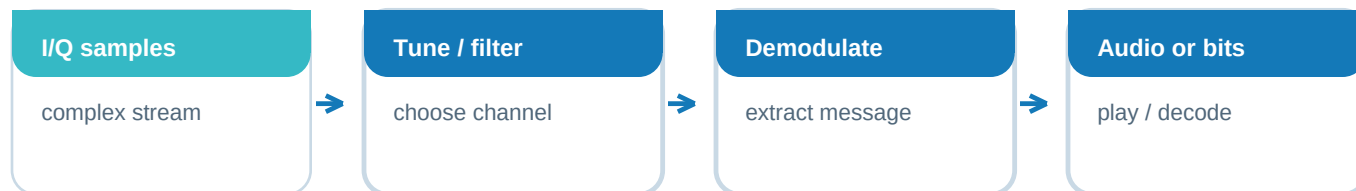
Each timestamp has a pair: $I[n]$, $Q[n]$. A sample rate of 2.4 MS/s produces 2.4 million pairs per second. These are usually signed 8-, 12-, or 16-bit values.

What software does with I/Q

Once captured, ordinary math turns a complex stream into stations, audio, bits, or pictures.

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3. The basic receive chain

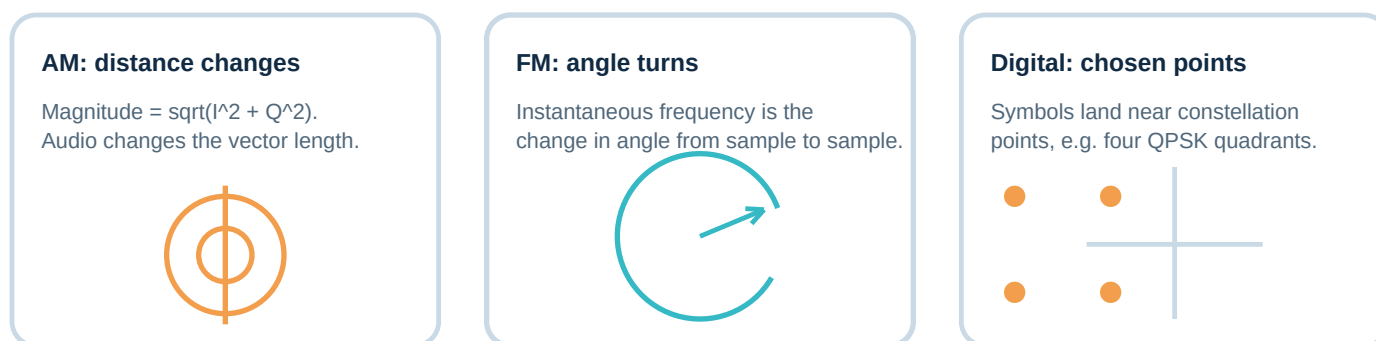


Software tuning is just rotation

To move a signal offset by f_0 down to zero, multiply every I/Q pair by a rotating reference. This shifts the desired station to baseband, where a narrow digital filter can isolate it.

$$\text{shifted}[n] = (I[n] + jQ[n]) \times \exp(-j 2 \pi f_0 n / F_s)$$

4. The message is encoded in how the point moves



5. A few practical mental models

Bandwidth costs samples

Your sample rate must cover the signal bandwidth. More rate sees more spectrum, but creates more data.

Gain affects dynamic range

Too little gain buries weak signals; too much clips I/Q values. Aim for headroom, not maximum levels.

I/Q imperfections are correctable

DC offset, gain mismatch, and phase mismatch distort the cloud of points. SDR software can estimate and remove much of this.

One-line takeaway

I/Q is a time series of 2D vectors. Preserve both axes, and software can tune, filter, and decode the radio signal.